Find Three Largest Numbers

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The problem

- Return the three largest values (sorted ascending) without sorting the input.

Intuition

- Keep a tiny sorted window of the three largest seen so far, and shift values into place as larger ones arrive.

Approach

- 1. Start with three empty slots
- 2. Compare each number with the largest slot first
- 3. Shift smaller values down to make room
- 4. Return the three values ascending

Java solution

```java
public class FindThreeLargestNumbers {
    public static int[] findThreeLargestNumbers(int[] array) {
        int[] top = {Integer.MIN_VALUE, Integer.MIN_VALUE, Integer.MIN_VALUE};
        for (int x : array) {
            if (x > top[2]) { top[0] = top[1]; top[1] = top[2]; top[2] = x; }
            else if (x > top[1]) { top[0] = top[1]; top[1] = x; }
            else if (x > top[0]) top[0] = x;
        }
        return top;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Keep a tiny sorted window of the three largest seen so far, and shift values into place as larger ones arrive.
- Complexity: O(n)